Table of Contents

..... 1
PARAMETERS 1
1. 8-BIT QUANTIZATION 1
2. WORD-LENGTH COMPARISON 2
3. DISPLAY QUANTIZATION RESULTS 2
4. FIRST 10 SAMPLES TABLE 3
5. WORD-LENGTH TABLE 3
6. OUTPUT PARAMETER TABLE 4
GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL 5
GRAPH 2: ROUNDING AND TRUNCATION ERROR 6
GRAPH 3: SQNR VS WORD LENGTH 6
GRAPH 4: QUANTIZATION ERROR VS WORD LENGTH 7
7. FINAL RESULT 8


```
clc;
clear;
close all;

format short g;
```

PARAMETERS

```
A = 1;
f = 50;
Fs = 1000;
N = 1000;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

B = 8;
bits = [4 8 12 16];
```

1. 8-BIT QUANTIZATION

```
D = 2*A/2^B;

% Rounding and truncation
xr = D*round(x/D);
xt = D*fix(x/D);

% Quantization errors
er = x - xr;
et = x - xt;

% SQNR calculations
SQr = 10*log10(mean(x.^2)/mean(er.^2));
```

```
SQt = 10*log10(mean(x.^2)/mean(et.^2));

% Output quantization
y = x;
yq = D*round(y/D);
eo = y-yq;

SQout = 10*log10(mean(y.^2)/mean(eo.^2));
```

2. WORD-LENGTH COMPARISON

```
step = zeros(size(bits));
sqnrR = zeros(size(bits));
sqnrT = zeros(size(bits));
maxER = zeros(size(bits));
maxET = zeros(size(bits));

for k = 1:length(bits)

    b = bits(k);
    d = 2*A/2^b;

    qr = d*round(x/d);
    qt = d*fix(x/d);

    er1 = x-qr;
    et1 = x-qt;

    step(k) = d;

    sqnrR(k) = 10*log10( ...
        mean(x.^2)/mean(er1.^2));

    sqnrT(k) = 10*log10( ...
        mean(x.^2)/mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));

end
```

3. DISPLAY QUANTIZATION RESULTS

```
fprintf('\n=========================================\n');
fprintf('          QUANTIZATION RESULTS\n');
fprintf('=========================================\n');

fprintf('Amplitude = %.1f | Frequency = %d Hz | Fs = %d Hz\n', ...
    A,f,Fs);

fprintf('Samples = %d | Main word length = %d bits\n',N,B);
fprintf('8-bit Step Size = %.8f\n',D);
fprintf('Rounding SQNR = %.4f dB\n',SQr);
```

```
fprintf('Truncation SQNR = %.4f dB\n',SQt);
fprintf('Output SQNR = %.4f dB\n',SQout);
fprintf('Max Rounding Error = %.6f\n',max(abs(er)));
fprintf('Max Truncation Error = %.6f\n',max(abs(et)));
```

```
==========================================
          QUANTIZATION RESULTS
==========================================
Amplitude = 1.0 | Frequency = 50 Hz | Fs = 1000 Hz
Samples = 1000 | Main word length = 8 bits
8-bit Step Size = 0.00781250
Rounding SQNR = 48.9339 dB
Truncation SQNR = 45.2931 dB
Output SQNR = 48.9339 dB
Max Rounding Error = 0.003483
Max Truncation Error = 0.005744
```

4. FIRST 10 SAMPLES TABLE

```
T1 = table(n(1:10)',x(1:10)',xr(1:10)', ...
    er(1:10)',xt(1:10)',et(1:10)', ...
    'VariableNames',{'n','Original','Rounded', ...
    'RoundError','Truncated','TruncError'});

fprintf('\nFIRST 10 QUANTIZATION VALUES\n');
disp(T1);
```

```
FIRST 10 QUANTIZATION VALUES
    n    Original    Rounded    RoundError    Truncated    TruncError
    _    ________    _______    __________    _________    __________

    0           0          0             0            0             0
    1     0.30902     0.3125     -0.003483      0.30469     0.0043295
    2     0.58779    0.58594     0.0018478      0.58594     0.0018478
    3     0.80902     0.8125     -0.003483      0.80469     0.0043295
    4     0.95106    0.95312    -0.0020685      0.94531      0.005744
    5           1          1             0            1             0
    6     0.95106    0.95312    -0.0020685      0.94531      0.005744
    7     0.80902     0.8125     -0.003483      0.80469     0.0043295
    8     0.58779    0.58594     0.0018478      0.58594     0.0018478
    9     0.30902     0.3125     -0.003483      0.30469     0.0043295
```

5. WORD-LENGTH TABLE

```
T2 = table(bits',step',sqnrR',sqnrT',maxER',maxET', ...
    'VariableNames',{'Bits','StepSize','SQNR_Round_dB', ...
    'SQNR_Trunc_dB','MaxError_Round','MaxError_Trunc'});

fprintf('\nWORD-LENGTH COMPARISON\n');
disp(T2);
```

```
WORD-LENGTH COMPARISON
    Bits     StepSize     SQNR_Round_dB    SQNR_Trunc_dB    MaxError_Round
MaxError_Trunc
    ____    __________    _____________    _____________    ______________
_____________

      4          0.125       23.667           20.871
0.059017          0.087785
      8       0.0078125      48.934           45.293
0.003483          0.005744
     12      0.00048828      78.811           65.891
0.00011536        0.00042324
     16      3.0518e-05      99.697           91.599
1.382e-05         2.6516e-05
```

6. OUTPUT PARAMETER TABLE

```
Parameter = {
    'Quantized signal'
    'Quantization error'
    'SQNR'
    'Word-length comparison'
};

Observation = {
    sprintf('8-bit rounding, Delta = %.7f',D)
    sprintf('Max rounding error = %.6f',max(abs(er)))
    sprintf('Rounding = %.4f dB; Truncation = %.4f dB',SQr,SQt)
    'SQNR increases as word length increases'
};

Unit = {
    'Amplitude'
    'Amplitude'
    'dB'
    'Bits / dB'
};

OutputTable = table(Parameter,Observation,Unit);

fprintf('\nOUTPUT PARAMETER TABLE\n');
disp(OutputTable);
```

```
OUTPUT PARAMETER TABLE
            Parameter
Observation                              Unit
    ____________________________
_____________________________________________________    ______________

    {'Quantized signal'      }    {'8-bit rounding, Delta =
```

```
0.0078125'              }    {'Amplitude'}
    {'Quantization error'    }    {'Max rounding error =
0.003483'                }    {'Amplitude'}
    {'SQNR'                  }    {'Rounding = 48.9339 dB; Truncation =
45.2931 dB'}    {'dB'       }
    {'Word-length comparison'}    {'SQNR increases as word length
increases'       }    {'Bits / dB'}
```

GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL

```
figure('Name','Original and 8-bit Quantized Signal');

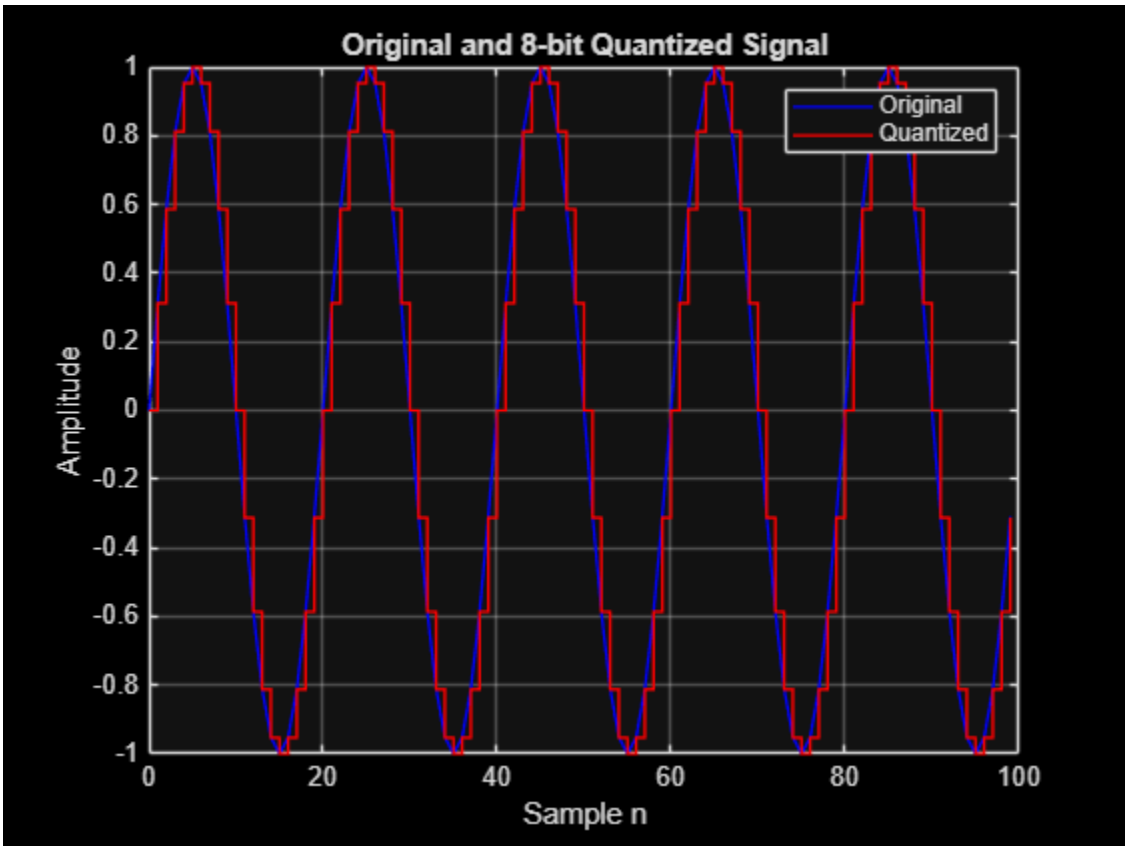
plot(n(1:100),x(1:100),'b-','LineWidth',1);
hold on;

stairs(n(1:100),xr(1:100),'r-','LineWidth',1);

grid on;
xlabel('Sample n');
ylabel('Amplitude');

title('Original and 8-bit Quantized Signal');

legend('Original','Quantized','Location','northeast');
```



GRAPH 2: ROUNDING AND TRUNCATION ER-ROR

```
figure('Name','Rounding and Truncation Quantization Error');

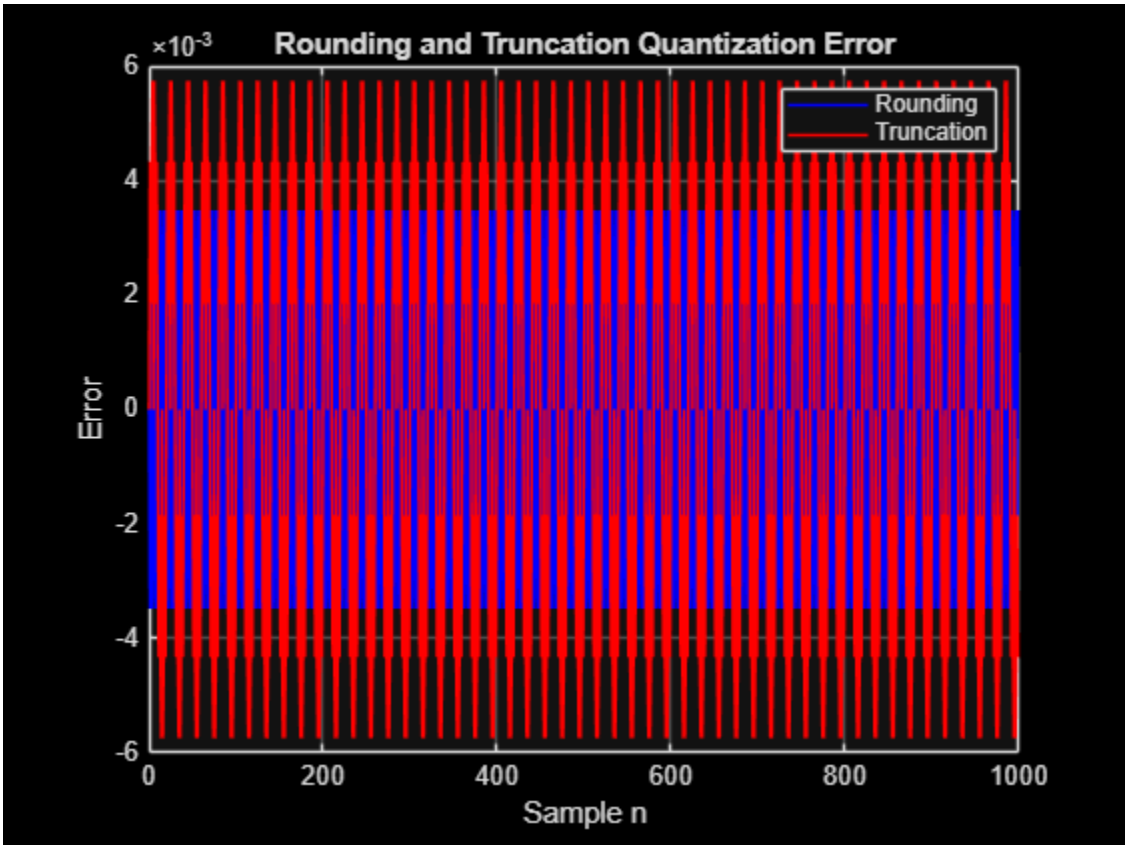
plot(n,er,'b-');
hold on;

plot(n,et,'r-');

grid on;
xlabel('Sample n');
ylabel('Error');

title('Rounding and Truncation Quantization Error');

legend('Rounding','Truncation','Location','northeast');
```



GRAPH 3: SQNR VS WORD LENGTH

```
figure('Name','SQNR vs Word Length');

plot(bits,sqnrR,'o-','LineWidth',1.2);
hold on;
```

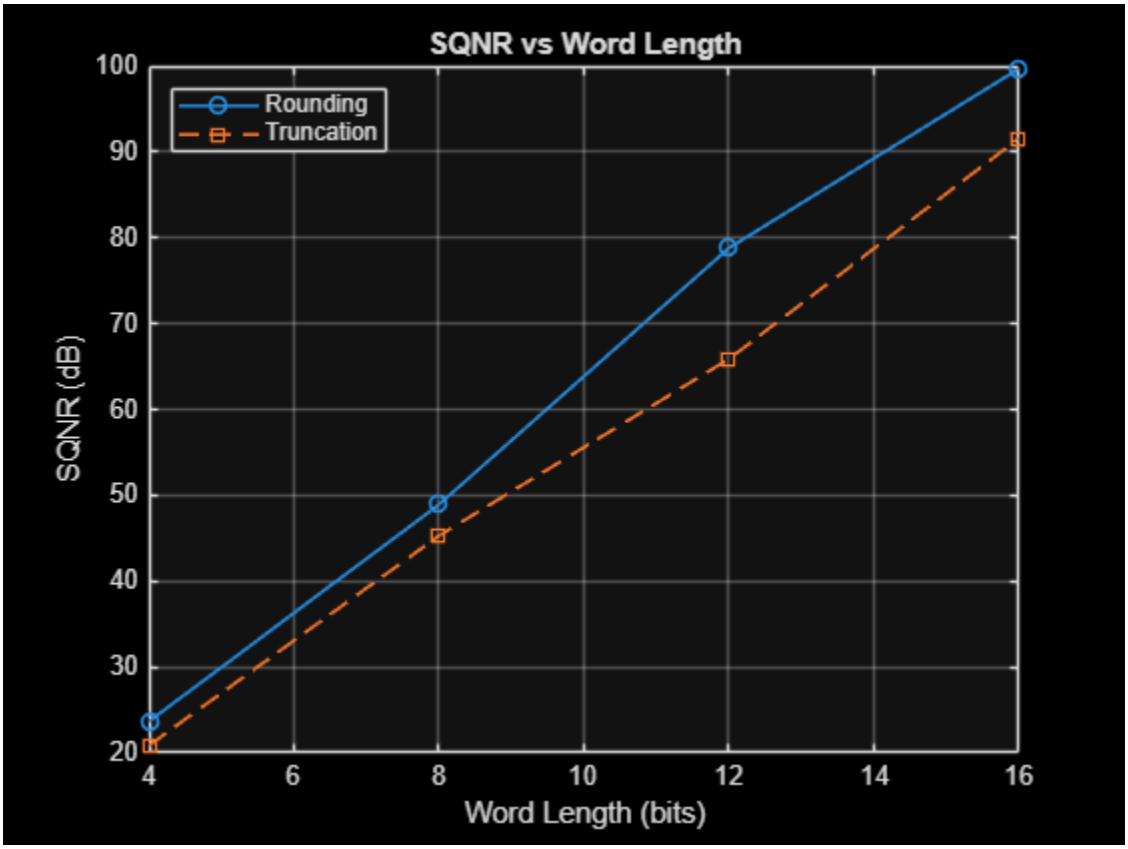
```
plot(bits,sqnrT,'s--','LineWidth',1.2);

grid on;
xlim([4 16]);

xlabel('Word Length (bits)');
ylabel('SQNR (dB)');

title('SQNR vs Word Length');

legend('Rounding','Truncation','Location','northwest');
```



GRAPH 4: QUANTIZATION ERROR VS WORD LENGTH

```
figure('Name','Quantization Error vs Word Length');

semilogy(bits,maxER,'o-','LineWidth',1.2);
hold on;

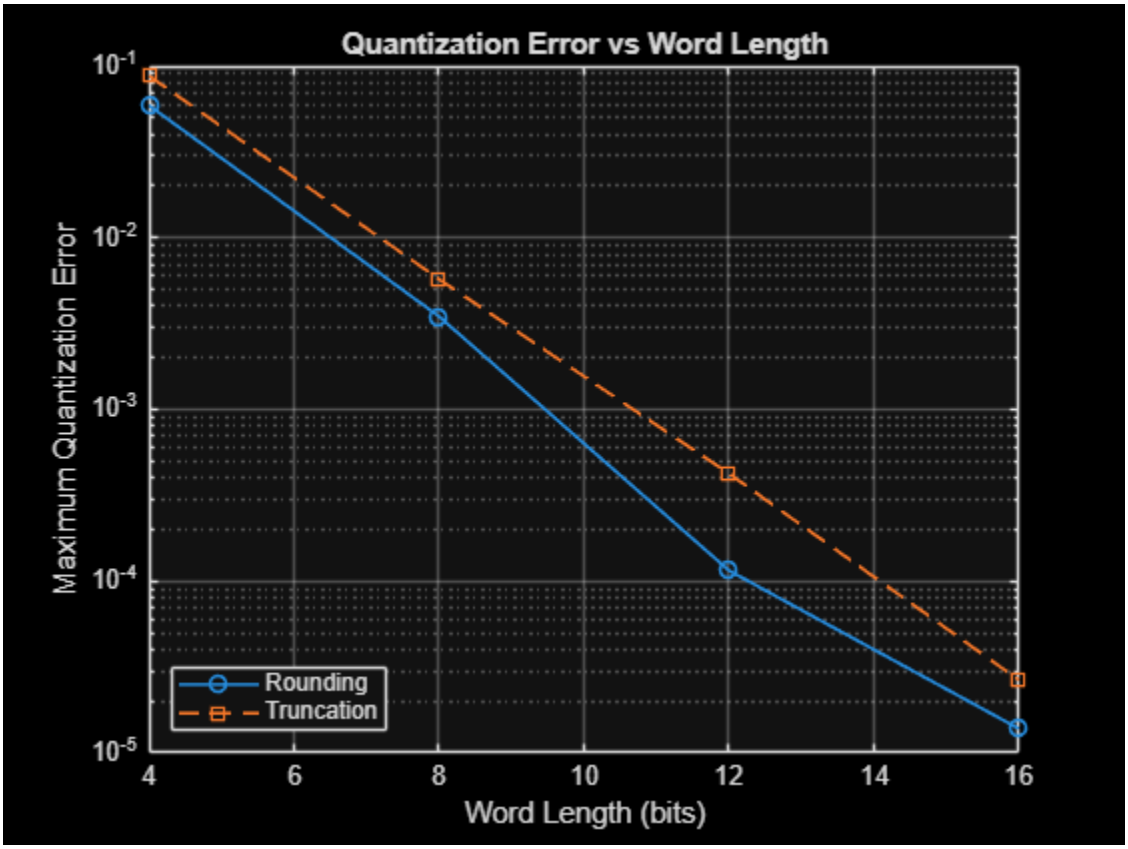
semilogy(bits,maxET,'s--','LineWidth',1.2);

grid on;
xlim([4 16]);
```

```
xlabel('Word Length (bits)');
ylabel('Maximum Quantization Error');

title('Quantization Error vs Word Length');

legend('Rounding','Truncation','Location','southwest');
```



7. FINAL RESULT

```
fprintf('\n=========================================\n');
fprintf('              FINAL RESULT\n');
fprintf('=========================================\n');

fprintf(['Uniform quantization using rounding and ' ...
    'truncation was performed.\n']);

fprintf('Quantization error and SQNR were calculated.\n');

fprintf(['Increasing word length reduces quantization ' ...
    'error and improves SQNR.\n']);

fprintf('Finite word-length effects are verified.\n');


=========================================
```

```
              FINAL RESULT
==========================================
Uniform quantization using rounding and truncation was performed.
Quantization error and SQNR were calculated.
Increasing word length reduces quantization error and improves SQNR.
Finite word-length effects are verified.
```

Published with MATLAB® R2026b